

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 03

Costing / Cost Estimation

Course	BECM 3101 - Construction and Project Management I
Lecture deck	3.0 - Costing (36 PDF pages/slides)
Question bank	Term-final examinations, 2016-2024 (25 pages)
Method	Every costing-related past question was matched to the primary and supporting lecture slides, irrespective of the original Section A/B label.

RESULT

This is a high-priority numerical and theory lecture. Sixteen past-question links were found. The most repeated blocks are stakeholder objectives, the builder's cost/bid-price structure and the refinery order-of-magnitude problem.

Prepared as part of the cumulative lecture-by-lecture question-bank analysis.

1. Executive Summary

The lecture progresses from objectives and cost categories to design/bid/control estimates, cost-estimation approaches, direct and joint-cost allocation, cost indices and screening estimates. The mapping below uses PDF page/slide numbers, counting the title page as Slide 1.

36	16	6	2	8	186
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Classification	Meaning
Exact	The requested content or numerical problem appears essentially as asked, with sufficient slide material for a complete answer.
Direct	The lecture provides the method and most answer content, but the student must adapt data, combine blocks or complete a small inference.
Partial	Only one component, formula or list is present; another lecture, textbook explanation or additional worked method is needed.

Slide-numbering note

The report uses PDF page/slide numbers. The title page is Slide 1; therefore the small printed number at the bottom of most lecture slides is one less. Example: PDF Slide 17 is printed as slide 16 in the deck.

Highest-priority slide blocks

Priority	Slides	Topic	Why it matters
1	27-33	Cost indices and refinery order-of-magnitude estimate	A full 15-20 mark numerical problem appears in 2016, 2019 and 2023. Learn the six adjustment steps and the final sequence.
2	3-5	Objectives from owner, designer and contractor perspectives	Appears in 2018 and as half of a repeated contract-plus-costing question in 2019, 2021, 2022 and 2023.
3	17-19, 23	Bid-price and construction-cost structure	Supports theory in 2017 and repeated average-bid-price questions in 2019-2021.
4	23-26	Direct cost and joint-cost allocation	The 2023 supervision/overhead allocation question is copied directly from Problem #1.

2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed task. 'Primary' identifies the best slide to begin from; 'Support' identifies the remaining slides needed for a complete response. 'QB p.' is the page of the uploaded question bank.

Examinations 2016-2019

Year	Original	Marks	QB p.	Question / task	Match	Primary	Support	Coverage assessment
2016	Sec B Q6(a)	10	2	How can cost estimation be approached? Briefly discuss the bid estimate.	Direct	17, 22	18-20	Slide 22 lists the four basic approaches. Slides 17-20 define the bid estimate, give the bid-price structure and identify direct, indirect and head-office cost components. The four approaches are named but not fully explained in the deck.
2016	Sec B Q6(b)	10	2	Write a short note on initial capital cost and state the objectives of cost estimation.	Exact	3, 7	4-5, 8	Slides 3-5 give the owner, designer and contractor objectives. Slides 7-8 list the initial capital-cost components. Both requested parts are directly available.
2016	Sec B Q8(b)	20	3	Prepare an order-of-magnitude estimate for a refinery after capacity, inflation, location, equipment and contingency adjustments.	Direct	30-33	27-29	The lecture contains the same refinery-estimation model, parameters and complete stepwise method. The examination changes the base/completion years, so the exact inflation exponent must follow the exam data.
2017	Sec B Q7(a)	7	5	From the builder's perspective, write down the basic structure of construction cost.	Exact	17	18-19, 23	Slide 17 gives the complete structure: bid price = total field cost + markup; total field cost = basic cost + field supervision; markup = head-office overhead + profit. Supporting cost components appear on Slides 18-19 and 23.
2018	Sec A Q1(b)	12	7	Write down the objectives of cost estimate from the owner, designer and contractor perspectives.	Exact	3-5	-	This question follows Slides 3, 4 and 5 almost point-for-point.
2019	Sec B Q5(a)	10	10	Define construction contract legally and state the objectives of cost estimation for different stakeholders.	Partial	3-5	-	Slides 3-5 fully cover the cost-estimation portion. The legal definition of construction contract belongs to Lecture 02.
2019	Sec B Q7(b)	10	10	Given field supervision, total/basic cost and a range of head-office overhead percentages, calculate the average bid price when profit is nil.	Partial	17	18-19	The bid-price formula and overhead categories are supplied. The averaging instruction and interpretation of the variables are not demonstrated as a worked example in this deck.
2019	Sec B Q7(c)	15	10-11	Make an order-of-magnitude estimate for the Gary-to-Los-Angeles refinery problem.	Exact	30-33	27-29	This is the lecture's Problem #2, including the same capacity rule, inflation, location indices, pollution equipment and contingency adjustment, followed by the complete solution.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2020-2024

Year	Original	Marks	QB p.	Question / task	Match	Primary	Support	Coverage assessment
2020	Sec B Q5(c)	10	13	Calculate the average bid price from field supervision cost, total/basic cost and minimum/maximum head-office overhead, with zero profit.	Partial	17	18-19	The required cost structure is shown on Slides 17-19. The exact average-overhead calculation is not worked in the slides.
2021	Sec B Q5(a)	8	16	Define construction contract legally and state the objectives of cost estimation for different stakeholders.	Partial	3-5	-	The cost-estimation objectives are directly covered; the legal contract definition is in Lecture 02.
2021	Sec B Q6(b)	12	16	Calculate the average bid price using field supervision cost and a 2%-4% head-office-overhead range, with nil profit.	Partial	17	18-19	Slides 17-19 supply the cost equation and overhead meaning, but not the complete worked calculation.
2022	Sec B Q5(a)	10	19	Define construction contract and describe the objectives of cost estimation from different stakeholder perspectives.	Partial	3-5	-	The stakeholder-objective component is exact; contract definition comes from Lecture 02.
2023	Sec B Q5(a)	10	21	Define construction contract legally and state the objectives of cost estimation for different stakeholders.	Partial	3-5	-	Slides 3-5 answer the costing half exactly. Use Lecture 02 for the legal contract definition.
2023	Sec B Q6(b)	13	21	Prorate a field supervision cost of \$13,245 and a 4% general-office overhead among formwork, rebars and concrete.	Exact	26	23-25	This is the same data and task as Problem #1. Slide 26 contains the worked allocation table; Slides 23-25 provide the formulas and cost bases.
2023	Sec B Q8(c)	16	22	Estimate the order of magnitude for the Gary-to-Los-Angeles refinery project.	Exact	30-33	27-29	The question reproduces the lecture's refinery problem and its adjustment sequence. The complete solution is on Slide 33.
2024	Sec B Q8(b)	13	25	Select and justify an appropriate cost-estimation approach using past-project data, a BOQ and detailed material, labour and equipment records; explain the other approaches.	Partial	22	11-20, 23	Slide 22 identifies production function, empirical cost inference, unit costs for BOQ and allocation of joint costs. Slides 17-20 and 23 support detailed bid/direct-cost estimation. The deck directs students to the reference textbook for full explanations of the four approaches.

3. Repeated Question Patterns

Repeated clusters are the fastest way to prioritize revision. The same content may move between Section A and Section B, so revise by topic rather than by section label.

#	Pattern	Past questions	Slides	Study implication
1	Stakeholder objectives of cost estimation	2018 Q1(b); 2019 Q5(a); 2021 Q5(a); 2022 Q5(a); 2023 Q5(a)	3-5	5 appearances. The 2018 question is pure costing; the other four combine it with the legal definition of a construction contract.
2	Basic cost / bid-price structure and overhead	2017 Q7(a); 2019 Q7(b); 2020 Q5(c); 2021 Q6(b)	17-19, 23	4 appearances. Memorize the equation and the direct/indirect/HQ-overhead components. The average-bid numerical step is only partially demonstrated.
3	Refinery order-of-magnitude estimate	2016 Q8(b); 2019 Q7(c); 2023 Q8(c)	27-33	3 appearances. This is the most important long numerical problem in the lecture.
4	Approaches to cost estimation	2016 Q6(a); 2024 Q8(b)	22	2 appearances. The slide names four approaches, but the reference textbook is required for full theory explanations.
5	Prorating field supervision and office overhead	2023 Q6(b)	23-26	1 exact appearance. The exam values are identical to the lecture problem.

Year coverage

Year	Links	Marks	Main coverage
2016	3	40	Approaches + bid estimate; initial capital cost/objectives; refinery problem
2017	1	7	Builder's construction-cost structure
2018	1	12	Stakeholder objectives
2019	3	35	Stakeholder objectives; average bid price; refinery problem
2020	1	10	Average bid price (partial support)
2021	2	20	Stakeholder objectives; average bid price
2022	1	10	Stakeholder objectives
2023	3	39	Stakeholder objectives; joint-cost allocation; refinery problem
2024	1	13	Selecting cost-estimation approaches

4. Slide-to-Question Reverse Index

Use this page while reading the lecture: each slide block shows exactly which past questions it can answer or support.

Slides	Content	Mapped questions	Priority
3-5	Objectives of cost estimate: owner, designer, contractor	2016 Q6(b); 2018 Q1(b); 2019 Q5(a); 2021 Q5(a); 2022 Q5(a); 2023 Q5(a)	Very high
6-10	Facility cost categories; initial capital, operation/maintenance and contingencies	2016 Q6(b)	Medium
11-16	Types of estimates; screening and conceptual/preliminary estimates	2024 Q8(b) (support)	Medium
17-20	Bid estimate and bid-price structure; direct, indirect and HQ overhead	2016 Q6(a); 2017 Q7(a); 2019 Q7(b); 2020 Q5(c); 2021 Q6(b); 2024 Q8(b)	Very high
21	Control estimate	2024 Q8(b) (support)	Low
22	Four approaches to cost estimation	2016 Q6(a); 2024 Q8(b)	High
23-26	Direct/basic cost and allocation of field supervision and overhead	2017 Q7(a); 2023 Q6(b); 2024 Q8(b) (support)	High
27-29	Cost indices, screening-estimate applications and procedure	2016 Q8(b); 2019 Q7(c); 2023 Q8(c)	Very high
30-33	Refinery Problem #2 and complete solution	2016 Q8(b); 2019 Q7(c); 2023 Q8(c)	Critical
34-35	Conceptual-estimate homework and accuracy-level chart	No direct 2016-2024 question identified	Low

Core formula to memorize

$$\text{Bid price} / \text{Project value} = \text{Basic cost} + \text{Field supervision cost} + \text{Head-office overhead} + \text{Profit}$$

The same relationship is also presented as: Bid price = Total field cost + Markup; Total field cost = Basic cost + Field supervision; Markup = HQ overhead + Profit.

5. Exam-focused Study Plan

Recommended order for revising this lecture before the term final:

Order	Slides	Action
Step 1	Slides 30-33	Solve the refinery problem without looking at the solution. Preserve the order: remove special item -> capacity -> inflation -> location -> new equipment -> contingency.
Step 2	Slides 3-5	Prepare three separate answer headings: owner's perspective, designer's perspective and contractor's perspective.
Step 3	Slides 17-19 and 23	Memorize the bid-price structure and classify every item as basic/direct, field supervision/indirect, HQ overhead or profit.
Step 4	Slides 23-26	Repeat Problem #1 and understand why supervision is allocated by basic cost and office overhead by total field cost.
Step 5	Slides 11-16 and 22	Differentiate design, bid and control estimates; screening vs conceptual; and learn the four estimation approaches.
Step 6	Slides 6-10 and 27-29	Revise cost categories, cost indices and the six screening-estimate adjustments as short-note material.

Important limitations of this lecture

Not fully covered here	What to use instead
Legal definition of construction contract	Lecture 02 - Construction Contracts
Full explanations of production function, empirical inference, unit-cost and joint-cost approaches	The slide explicitly directs students to the Chris Hendrickson reference textbook.
Competitive-bidding probability, mark-up strategy and zero-profit rationale	Pricing and Bidding lecture
Average-bid-price worked example using minimum and maximum HQ-overhead percentages	Pricing/Bidding notes or instructor's worked solution; this lecture supplies only the base equation.
Detailed conceptual-estimate worked example	Slide 34 assigns it as homework rather than presenting the solution.

Bottom line

For maximum marks, prioritize Slides 3-5, 17-19, 23-26 and 27-33. Together they account for nearly every strong match identified in the 2016-2024 question bank.